

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Software engineer with over **5+ years** of hands-on experience delivering and validating **production-ready** systems in **real-world production environments** across SaaS and healthcare, backed by an **MS in Computer Science** and daily work in **Python** and **SQL**.
- Strong foundations in **software engineering**, **data structures** and applied **statistics**, applied to large-scale platform quality, data validation and measurement of system behaviour under real production load.
- Builds and operates automation systems **end-to-end** — from experimentation and **evaluation** through **deployment**, **monitoring** and iteration — using **CI/CD pipelines** in TeamCity, Jenkins and GitLab.
- Hands-on with **cloud environments**, including **AWS Lambda** serverless workloads, cloud-native service validation and **deploying services at scale** within regulated healthcare and SaaS estates.
- **Proficiency in Python** for automation frameworks, data validation scripts, test-data generation and API-driven workflows, applying **production-grade engineering practices** such as code review, version control and modular design.
- Designs **evaluation frameworks** and regression baselines that quantify system **reliability** and catch regressions before release — the same measurement discipline that underpins **model monitoring**.
- Improves **performance**, **cost efficiency** and **reliability** through experimentation and **tuning**, using JMeter load profiling and **observability** data to locate bottlenecks in distributed services.
- Collaborates with **cross-functional partners** — **product managers**, **platform engineers** and **data teams** — to translate complex business problems into working, **customer-facing** software.
- Works comfortably in **ambiguous or evolving problem spaces**, taking loosely specified requirements through exploration, prototyping and delivery inside two-week **Agile** sprints.
- Experienced with **MLOps**-adjacent tooling: **CI/CD pipelines**, artifact versioning, automated gating, environment promotion and **monitoring** of deployed services across staging and production.
- Deep **data exploration** and validation background using **SQL**, Postgres, MySQL and Oracle — profiling datasets, reconciling records and verifying the data integrity that feeds downstream analytics.
- **Strong problem-solving skills** demonstrated by isolating, reproducing and driving defects to resolution across n-tier, mobile and web architectures with heavy cross-team dependency.
- Applies **prompt engineering** and **LLM**-assisted tooling to accelerate test generation, code review and documentation workflows in day-to-day engineering practice.
- Hands-on **API** engineering and validation with REST, **Postman**, RESTAssured and Swagger — the same interface surface through which **AI models** are served and consumed in production.
- Comfortable across **Python**, C#, Java, TypeScript and SQL, moving between languages as platform and data teams require rather than specialising narrowly.
- Committed to **best practices** and **operational excellence**: reusable frameworks, documented standards, reproducible environments and measurable quality gates adopted across teams.
- Certified **Oracle Associate** and **ISTQB** professional with a consistent record of shipping reliable, **scalable** software in **Agile**, product-oriented organisations.

EDUCATION

University of Missouri Kansas City

“Master of Science” in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

“Bachelor of Technology” in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming Languages:** Python, C#, Java, SQL, TypeScript, JavaScript.
- **Artificial Intelligence & Machine Learning:** Machine learning, artificial intelligence, deep learning, PyTorch, TensorFlow, JAX, scikit-learn, Hugging Face Transformers, LangChain.
- **Generative AI & LLM Techniques:** Large language models (LLMs), generative AI, prompt engineering, retrieval-augmented generation (RAG), RAG pipelines, fine-tuning foundation models, embeddings, vector search, agentic workflows, agent-based architectures.
- **Machine Learning Foundations:** Statistics, data structures, algorithms, data modeling, feature engineering, model evaluation, model performance tuning, experimentation and A/B testing.
- **MLOps & Model Operations:** MLOps tooling, CI/CD pipelines, model monitoring, feature stores, experiment tracking, artifact versioning, automated deployment gating, intelligent automation.
- **Cloud Platforms:** AWS (Lambda, S3, CloudWatch), Azure, GCP, serverless deployment, deploying services at scale.
- **Data & Databases:** Postgres, MySQL, Oracle, Vertica, SQL query optimisation, data exploration, data quality validation.
- **CI/CD & DevOps:** Jenkins, TeamCity, GitLab CI, Circle CI, Git, GitHub, Bitbucket, Maven, Yarn, NPM.
- **Observability & Monitoring:** Splunk, CloudWatch, structured logging, metrics dashboards, alerting, performance profiling.
- **API & Services:** REST, RESTAssured, Postman, SoapUI, Swagger/OpenAPI, Dotnet Core 6.0 Web API, JSON.
- **Testing & Evaluation Frameworks:** PyTest, JUnit, TestNG, XUnit, JEST, Cucumber/BDD, Page Object Model, Data Driven, Keyword Driven.
- **Automation Tools:** Selenium WebDriver, Playwright, Appium, Puppeteer, Lambda testing tool, Sause labs.
- **Performance & Reliability:** JMeter, load and stress profiling, latency analysis, cost efficiency tuning, reliability engineering.
- **Web Technologies:** Dotnet Core, HTML/HTML5, CSS2/CSS3, JavaScript, TypeScript, Angular, React JS.
- **Tooling & Collaboration:** Jira, Azure DevOps, Confluence, TestRail, Bugzilla, VS Code, PyCharm, IntelliJ, Eclipse, Visual Studio.

EXPERIENCE

Test Automation Engineer

Oct 2024 – Present

PCMI LLC (Full Time)

Chicago, IL, USA

- Designed, developed and deployed a **Python**- and **C#**-based automation platform for PCRS, PCMI's **enterprise SaaS** product, applying **production-grade engineering practices** across modular design, peer review and Git version control.
- Built and operated validation systems **end-to-end** — experimentation and **evaluation** through **deployment**, **monitoring** and iteration — so **reliability** regressions surface before reaching **customer-facing** releases.
- Deployed automated suites into **CI/CD pipelines** on TeamCity, gating every release behind reproducible checks and giving the **organization** rapid feedback on **scalable**, **production-ready** delivery.
- Improved **performance**, **cost efficiency** and **reliability** through parallel execution and selective regression, cutting feedback latency while holding coverage steady as the suite kept **scaling**.
- Instrumented **observability** across automated runs — structured logging, failure classification, metric dashboards — converting raw output into actionable signal for engineering.
- Designed **evaluation frameworks** and regression baselines quantifying platform behaviour release over release, the same measurement discipline that underpins **model monitoring** in production systems.
- Collaborated closely with **product managers**, **platform engineers** and **data teams** to translate complex **business problems** into specifications embedded in **internal platforms** and client-facing **products**.
- Performed **data exploration** and reconciliation in **SQL** across relational schemas, validating the datasets that feed downstream reporting and **data-driven decision making**.

- Operated in an **ambiguous and evolving problem space**, taking loosely specified insurance-domain requirements from exploration through prototype to verified delivery.
- Validated **enterprise SaaS** releases across system integration, UAT and production verification, covering **real-world** usage paths through business-**critical** warranty workflows.
- Established **best practices** for automation architecture, reusable utilities and environment promotion, evolving how the group builds and operates **production-ready capabilities**.
- Diagnosed **critical** production defects alongside development teams, reproducing, isolating root cause and driving each to resolution against active release timelines.

Environments: Python, C#, Selenium WebDriver, Playwright, TeamCity, CI/CD, SQL, Postgres, Git, Jira, REST APIs, Agile/Scrum, UAT, System Integration Testing, Observability, Test Data Management, SaaS Platform Testing, Enterprise Platform Validation.

SDET / QA Automation Engineer

Dec 2023 – Sept 2024

Client: CVS

Remote, USA

- Designed, developed and deployed **Python**, C# and TypeScript automation for **cloud-based** healthcare platforms, applying **production-grade engineering practices** across a distributed **enterprise organization**.
- Deployed and validated **cloud-native** services on **AWS Lambda**, exercising serverless workloads for correctness, cold-start latency and cost behaviour while **deploying services at scale**.
- Built **API-level** verification for REST services with **Postman**, RESTAssured, Swagger and XUnit — the interface layer through which platform **capabilities** and **models** are served and consumed.
- Embedded automated suites into **CI/CD pipelines** on GitLab, enforcing quality gates on every merge and shortening the path from commit to verified **deployment**.
- Ran **performance experimentation** with JMeter across load profiles, quantifying latency, throughput and **cost efficiency** to guide capacity and **tuning** decisions.
- Monitored deployed services through Splunk dashboards and structured logging, closing the loop from **deployment** back into **monitoring** and iteration.
- Collaborating closely with **product** and **platform teams**, converted ambiguous clinical requirements into verifiable acceptance criteria for **customer-facing** healthcare **products**.
- Applied **Python** for synthetic **data exploration** and test-data generation, producing **real-world-shaped** datasets while preserving PHI safety and **responsible** data-handling standards.
- Designed **evaluation frameworks** with JUnit, TestNG and XUnit that scored release readiness against measurable criteria rather than subjective sign-off.
- Automated regression suites end-to-end with Playwright and Selenium WebDriver, reducing manual cycles and freeing engineering time for exploratory and **experimentation** work.
- Contributed to release tooling — artifact versioning, environment promotion and automated rollback — improving **reliability** across a frequent release cadence.
- Validated data moving between **enterprise** healthcare services using **SQL**, confirming schema-level integrity across MySQL-backed **internal platforms**.
- Practised **BDD** with Cucumber and Gherkin so cross-functional partners could read, review and contribute to executable specifications directly.
- Investigated **critical** production incidents across **cloud** and on-premise tiers, reproducing failures, isolating root cause and driving fixes with the owning teams.
- Operated within **Agile** ceremonies alongside product, engineering and **data teams**, adapting scope as programme priorities shifted through delivery.
- Documented framework architecture and onboarding guides, growing **technical** capability across the team and reducing ramp time for new engineers.

Environments: Python · C# · TypeScript · .NET · Playwright · Selenium WebDriver · AWS Lambda · Amazon Web Services (AWS) · Cucumber · Gherkin · Behavior-Driven Development (BDD) · JUnit · TestNG · XUnit · Postman API · RESTAssured · Swagger API · REST APIs · GitLab · CI/CD · JMeter · Splunk · MySQL · Git · Jira · TestRail · Agile/Scrum

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Developed **scalable** automation in Java and **Python** on a **Page Object Model** architecture, raising test case development to 90% while keeping the framework maintainable as the team grew.

- Deployed mobile automation with Appium and SeeTest across iOS and Android, contributing to a 95% improvement in measured product health against stated requirements.
- Delivered **API** verification with **Postman** and SoapUI across REST and SOAP surfaces, reaching roughly 90% coverage of the **enterprise** service contract.
- Integrated automated runs into Jenkins **CI/CD pipelines**, moving the team from manual triggers to continuous scheduled execution against every build.
- Applied the Cucumber **BDD** framework to express behaviour as readable specifications shared directly with **product managers** and **software engineers**.
- Used SQL for **data exploration** and backend validation, reconciling application state against persisted records across n-tier **enterprise** services.
- Developed, executed and maintained test plans, scripts and **evaluation** criteria, ensuring measurable quality objectives were met on every release.
- Worked in the XTEND high-level language on platform components, reaching 80% proficiency in an unfamiliar and **evolving problem space**.
- Participated in 80% of **product** design reviews, shaping **capabilities** before implementation began and reducing rework downstream.
- Collaborating closely with **cross-functional** teams through two-week sprints and monthly releases, carried user stories through to verified **real-world** outcomes.
- Ran regression cycles protecting existing functionality as the platform evolved, catching **reliability** regressions early in each cycle.
- Tracked and reported QA metrics and KPIs, giving engineering leadership **data-driven** visibility into release quality trends over time.
- Reproduced and isolated reported defects, tracking each through to resolution and feeding findings back into design and implementation.
- Partnered with teammates on higher-risk components and interfaces, building shared understanding of **technical** constraints and system dependencies.

Environments: Java · Python · Selenium WebDriver · Appium · SeeTest · Cucumber · Behavior-Driven Development (BDD) · XTEND · TestNG · Jest · Android Studio · Jenkins · CI/CD · Postman · SoapUI · REST APIs · SQL · Jira · QTest · TestRail · Test Planning · System Testing · Page Object Model · Agile/Scrum

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed **BDD** automation with Selenium WebDriver and Cucumber in Java, establishing the foundation of the team's regression capability at 85% hands-on ownership.
- Built mobile automation with Appium in Java, extending coverage across device platforms and reporting defects with reproducible, **real-world** evidence.
- Contributed to 80% of **CI/CD** pipeline tasks and 90% of **Agile** projects and Scrum ceremonies across the delivery portfolio.
- Practised functional, integration, system, regression and unit testing across an n-tier architecture, building broad **software engineering** and **data structures** foundations.
- Owned the defect lifecycle — writing and executing cases, logging, tracking and prioritising issues through to closure against release deadlines.
- Produced test reports summarising findings, resolutions and recommendations, giving engineering and delivery stakeholders **data-driven** decision support.
- Ran regression cycles verifying that new changes preserved existing behaviour release over release across the **enterprise** application estate.
- Developed working knowledge covering 75% of the **technical** stack in n-tier applications, from UI through services to database.

Environments: Java, J2EE, Selenium WebDriver, Cucumber, BDD, Appium, API Testing, System Integration Testing, UAT, Production testing, Agile/Scrum, Jira.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.